

Assessing the impact of socio-demographics and farming activity on ward-level mortality patterns using farm and population decennial censuses

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[Correction added on 29 May 2024,
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John Mallett was designated as first
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Abstract

Introduction and Objective: Farmers experience a specific set of unique dangers, which increases their risk of mortality compared with any other occupation. This study hypothesised that Northern Ireland's (NIs) agriculturally saturated Wards have a higher risk of mortality compared against non-agriculturally based Wards.

Design: The Population Census and Farm Census information were downloaded from the Northern Ireland Neighbourhood Service (NINIS) online depository to compile three mortality-based data sets (2001, 2011, pooled data sets). Assessing the impact of socio-demographics and farming activity on Ward-level mortality patterns using farm and population decennial censuses. This study analysed all 582 Ward areas of NI, which enclosed the entire populace of the country in 2001 and 2011.

Findings: Path analysis was utilised to examine direct and indirect paths linked with mortality within two census years (2001; 2011), alongside testing pathways for invariance between census years (pooled data set). Ward-level results provided evidence for exogenous variables to mortality operating through three/four endogenous variables via: (i) *direct effects* (age), (ii) *summed indirect effects* (age; males; living alone; farming profit; and deprivation) and (iii) *total effects* (age; males; living alone; and deprivation). Multi-group results cross-validated these cause-and-effect relationships relating to mortality.

Discussion and Conclusion: This study demonstrated that *farming intensity scores, farming profits and socio-demographics'* influence on *mortality risk* in a Ward were dependent on the specific social-environmental characteristics within that area. In line with earlier area level research, results support the aggregated interpretation that higher levels of farming activity within a Ward increase the risk of mortality within those Wards of NI. This was an essential study to enable

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future tailoring of new strategies and upgrading of current policies to bring about significant mortality risk change at local level.

KEYWORDS

aggregated data, agricultural areas, mortality patterns, NINIS data, path analysis

1 | INTRODUCTION

Farmers experience a specific set of unique dangers with greater exposure to fatal injury through workplace and environmental sources of risk, compared with any other occupation.^{1,2} This study has arisen in response to mixed findings^{3,4} and a lack of research on the relationship between farming, community characteristics and risk of mortality between areas with and without agricultural activity. This lack of research may be because nationally linked farm census and population census data at Ward level have not previously been available for NI.⁵ In addition, a better understanding is needed of how farmers' unique socio-environmental attributes may have multiplicative effects contributing to their risk of mortality at Ward-level in NI.

Earlier research has reported the agricultural industry is dominated by male workers due to the strenuous and onerous physical nature of their job.³ Two longitudinal US studies on the Alameda County data investigated high poverty neighbourhoods (e.g., low income, low education and unskilled male labourers.) on all-cause mortality risk. High poverty areas were utilised as a proxy measure for low socioeconomic status (SES) neighbourhoods, relating to ones social and physical environment. Overall findings indicated high poverty areas had almost twice the risk of mortality compared against affluent areas.^{6,7}

This is in line with the Republic of Ireland's (ROIs) national statistics relating to mortality differentials reporting that mortality rates at area level were dependent on that areas proportion of social occupations, lifestyle, highest level of education obtained and its deprivation ranking.⁸ This report concluded that the 2011 life expectancy at birth for ROI males living in the most deprived areas died 4.3years earlier (73.7years) compared with males living in the most affluent areas (78years). Likewise, an equivalent NI (2012–2018) report concluded that the life expectancy at birth between the most and least deprived areas remained almost twice as high in age gap for males (7.1years) compared with females (4.4years).⁹

Previous research on farmers health around the world have utilised 65+ year olds within an area to represent agricultural workers of retirement age continuing to farm. Lee and collaborators¹⁰ concluded that older aged farmers continue working on-farms with livestock and

What is already known on this subject

1. Farmers experience a specific set of unique dangers, which increases their risk of mortality compared to any other occupation.
2. Area-level studies using aggregated data suffer from a degree of unavoidable ecological bias, making researchers reluctant to analyse clustered data as there is a misconception their results have restricted power.
3. This study has arisen in response to mixed findings and a lack of research on the relationship between farming, community characteristics and risk of mortality between areas with/without agricultural activity.

What this paper adds

1. This papers farm census to population census linkage has provided a contextual mechanism for investigating and understanding some of the unexplained variations in mortality risk at Ward-level in NI, which is suitable for future researchers to use for international comparisons.
2. This study demonstrated that *farming intensity scores, farming profits, socio-demographics*' influence on *mortality* risk in a Ward was dependent on the specific social-environmental characteristics within that area.
3. Results indicated higher levels of farming activity within a Ward increases the risk of poor health and mortality within those Wards in 2001 and 2011.
4. This was an essential study to enable future tailoring of new strategies and upgrading of current policies to bring about significant mortality risk change at local level.

heavy machinery despite mobility issues, previous injuries or poor health, which further increases their risk of mortality. Likewise, Chan and colleagues¹¹ investigated socio-demographics on risk of mortality (1998–2002) between 458 counties. They found greater fatality rates than

expected in counties with higher percentages of: 65+ year olds, below degree qualifications, near poverty level incomes, lower county SES and higher amounts of farming activity (e.g., farming, fishing or forestry).

Korda and associates¹² investigated education-based inequalities (25–84 years) in all-cause mortality on mortality registrations linked to censuses (2011–2012); using age groups (25–84 years), sex and area-inequalities within Australia. They found males with below degree education were RR 2.2 [2.08–2.33] and females were RR 1.64 [1.55–1.74] times higher risk of mortality compared with their matched counterparts with above degree qualifications. Similarly, a study by Chetty and collaborators¹³ investigated the level, temporal trend and geographic variability between income (from tax records) and life expectancy at 40 years old using mortality records ($n = 1.4$ billion) in the United States (2001–2014). They concluded that higher income groups were associated with lower mortality rates within areas. These educational and income findings are supported by previous social health gradient research.^{6–8,14}

Norman and Purdham¹⁵ analysed variations in unpaid care provision within samples of anonymised records from the 2001 Census across 174 areas of England and Wales. They found the proportion of limiting long-term illnesses (LLTIs) within an area had strong predictive power with that areas proportion of unpaid carers. In contrast, a Japanese study by Miyawaki and collaborators¹⁶ found informal caregiving was linked with mental and physical burden. They concluded that female caregivers (aged 44–77 years) positioned within the lower SES groups were HR 1.8 [1.03–3] times higher risk of mortality than males between 2000 and 2010. Bentham and colleagues¹⁷ investigated LLTIs and their association with mortality and social deprivation ($n = 401$ local authority districts) in England and Wales. They concluded LLTIs and mortality were both positively correlated with social deprivation, while LLTIs had the strongest correlation with the elderly people (65+ year olds). Carson and associates⁹ report on NI summary statistics dovetail with these deprivation results.

Earlier research has found working over full-time hours weekly long-term causes poor work-life balance and increases the risk of various occupational health problems¹⁸ and mortality.^{4,19,20,21} Kandler and associates²² investigated the relationship between living alone, cardiovascular risk factors and mortality (45–74 years) on the population-based MONICA/KORA cohort ($n = 7016$) study (1984–1995) in Germany. After adjusting for socio-demographics, cardiovascular risk factors, and health behaviours, they found men living alone were HR 1.96 [1.56–2.46] times higher risk of mortality (however this risk was not observed in women). They concluded living alone was a strong psychosocial risk factor of mortality presumably due to these males having lower education,

fewer social networks and a heavier nicotine consumption, in comparison with their matched cohabitating counterparts.

Structural Equation Modelling (SEM) techniques were utilised in this study as they are effective and efficient at handling factor scores and mediation effects, which were analysed at single- and multiple-time points.^{23–27} Path analyses allowed the researchers to test new hypotheses, specify relationships amongst variables, while quantifying the strength and the direction of the cause-and-effect relationships.^{28–31} Extending SEM, multiple group analysis was used to assess whether and how populations differ through direct, indirect, and total path comparability over a 10-year period.^{27,32} Multi-group analysis was utilised, which allowed the researchers to test for similarities (invariance) and differences (non-invariance) between identical models within different populations (groups).³³ Thus enabling cross validation of results (2001, 2011), while ensuring they were reliable and replicable.^{27,30}

A socio-ecological model was used to frame conceptual causal pathways leading to mortality, to help understand the multifaceted additive/multiplicative effects of social, economic and environmental factors on *death rates* within urban and rural Wards.^{34,35} This study expanded on the design, methodology and analytic limitations of a neighbourhood social environment study conducted by Yen and Kaplan.⁷ The current study employed an interdisciplinary approach by merging Population Census with Farm Census information to address Ward-level determinants of mortality in agricultural Wards. The concept of farming intensity scores was utilised to assess the level of agricultural activity at Ward-level based on specific farm census data; farmers, grassland, farms, cattle, pigs and poultry.^{33,36,37}

Within this study, the factor of *Farming Intensity* was computed as a composite factor score for each Ward using six indicator items (farms, farmers, grass, cattle, pigs and poultry). *Farming Intensity* was the sum of different farm-related measures. An *Agricultural Ward* was defined as the area having: 1+ farm holdings; 1+ farmers; 1% or more grass for farming; and/or small, medium, large quantities of each animal (cattle, pigs, poultry) per Ward. In contrast, a *Non-Agricultural Ward* was defined as the area having: 0 farm holdings; 0 farmers; 0% farming grass; and/or no quantities of each animal per Ward.

Northern Ireland Statistics and Research Agency (NISRA) classified areas in 2015 as: 9.6% ($n = 56$) mixed, 54% ($n = 314$) urban and 36.4% ($n = 212$) of rural wards ($n = 582$).³⁸ In line with NISRA's classification, this paper considers Ward areas with less than 4500 residents as rural Wards and Wards with agricultural activity to be lower SES than their counterparts.^{38,39} In line with empirical research, this study utilised the Average Standard Gross Margin of Farms as a measure of on-farm livelihoods (or

Ward-level farming profit). Farm profit was calculated using the farm business revenue minus costs of farm activities equating to profits at Ward-level. This study speculated that the more farming profit earned within a Ward can be interpreted as a Ward with higher farming intensity scores and more agricultural affluence; compared to Wards with lower farming profit.^{14,38}

Figure 1 describes the proposed structure of the current path model analysed at Ward-level in 2001, with the analysis repeated in 2011. This path diagram provides a synopsis of where the seven exogenous and three/four endogenous variables were positioned within their direct and mediation paths linking to mortality across comparison groups (2001 and 2011 Wards). This study's objective was to investigate the additive and multiplicative effects of the exogenous variables on risk of mortality at Ward level both years.

The current study hypothesised that:

- Wards with greater amounts of *natural log of the population*, *65 to 100+ year olds*, *below degree qualifications* and lower *farming profits* will be associated with higher risk of *mortality* both years (direct effects).
- Wards with higher amounts of *natural log of the population* and *below degree qualifications* operating through

LLTIs, *unpaid carers*, and *full-time hours* will be associated with higher risk of *mortality* both years (sum of indirect and total effects).

- Wards with larger amounts of *65 to 100+ year olds*, *males*, and *living alone* operating through *LLTIs*, *unpaid carers*, *full-time hours*, and *farming intensity scores* will be associated with higher risk of *mortality* both years (sum of indirect and total effects).
- Wards with a high *deprivation rankings* (least deprived Wards) and high *farming profits* operating through *LLTIs*, *unpaid carers*, *full-time hours*, and *farming intensity scores* will be associated with lower *mortality* risk both years (sum of indirect and total effects).
- Constrained path equations are operating equally in a conceptually similar manner across 2001 and 2011 groups (multiple group analysis).

2 | METHODS

2.1 | Context of study

The Population Census and Farm Census information were downloaded from the Northern Ireland

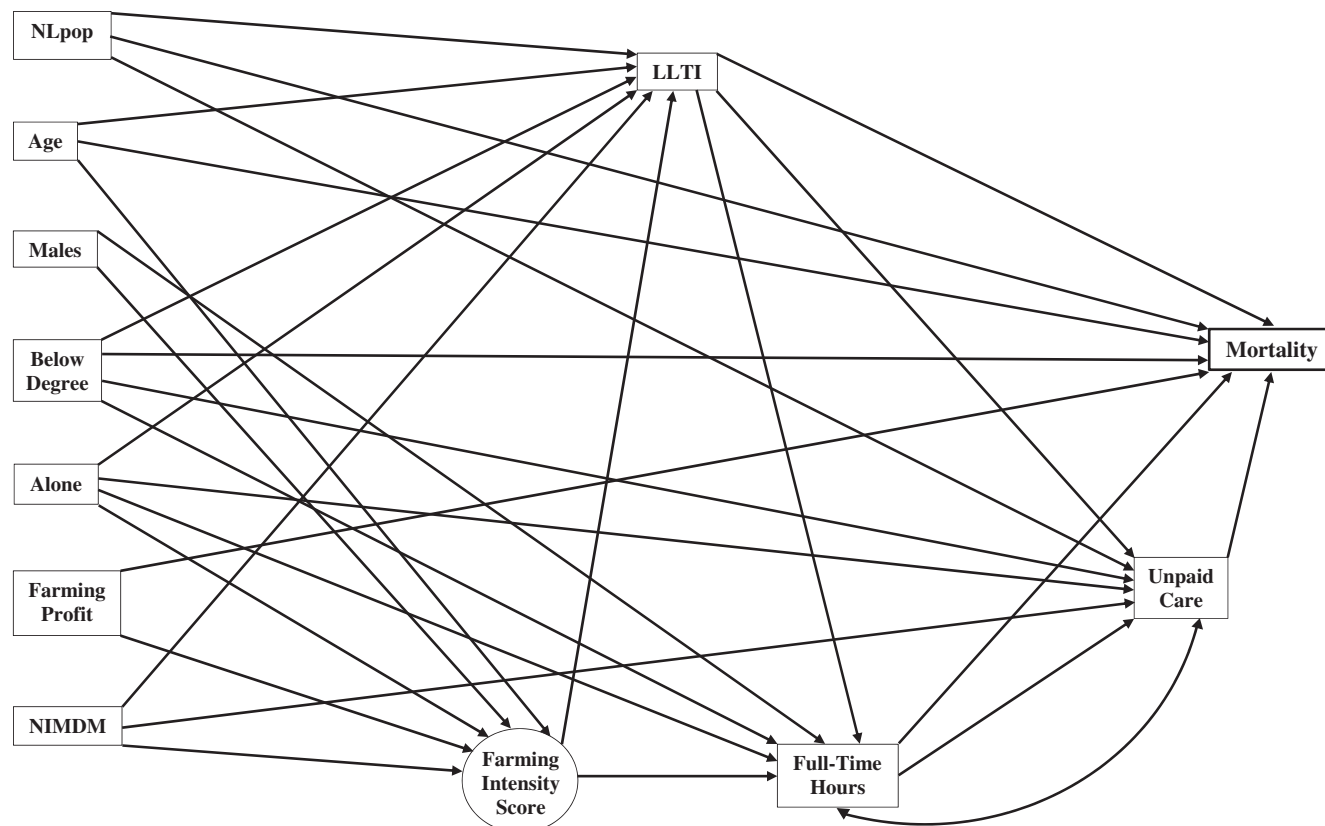


FIGURE 1 Path model of seven exogenous variables operating through three/four endogenous variables linking to mortality at Ward-level in 2001 and 2011. NLpop represents natural log of the population, which is utilised here as an offset variable (exposure). NIMDM represents the Rank of Northern Ireland Multiple Deprivation Measure. LLTI represents limiting long-term illness.

Neighbourhood Information Service (NINIS) online warehouse³⁸ and merged using SPSS-25⁴⁰ to build the 2001, 2011, and pooled data sets. These data were publicly available and ethics approval was granted in January 2019.

2.2 | Measures

The variables included in the model were operationalised in an identical way across 582 Wards within 2001 and 2011 data sets.

Deaths (mortality). The outcome variable represented the total number of *all-cause mortalities* recorded over a 12-month period per Ward (2001; 2011) and remained as a count.

Natural log of the population (NLpop) was calculated from the *Ward population* variable, then added to the model as a covariate.

Age represented the proportion of people 65 to 100+ years old in each Ward.

Gender represented the proportion of *males* per Ward.

Marital Status represented the proportion of residents aged 16+ *living alone* in each Ward.

Qualifications represented the proportion of residents aged 16+ with *below degree* level qualification(s) per Ward.

Limiting Long-Term Illness (LLTI) represented the proportion of residents who reported having a *limiting long-term illness* or disability in each Ward.

Unpaid Carers represented the proportion of residents aged 16+ who were providing 1+ hours of *unpaid care* weekly per Ward.

Hours Worked Weekly represented the proportion of residents aged 16+ recorded as working *full-time* in each Ward.

Rank of Northern Ireland Multiple Deprivation Measure (NIMDM) reflected the overall weighed deprivation score across seven domains; each of the 582 Wards were ranked from most deprived (1) to least deprived (582).

Average Standard Gross Margin (SGM) represented the annual *farming profit* earned from arable and/or livestock production (1-unit equates to €1000) per Ward.

Farms represented the total number of active farms per Ward and remained as a count.

Agricultural Labour (farmers) represented the proportion of active *farmers* as a percentage of each Ward population aged over 16 years old.

Grass represented the proportion of *grass* in hectares relative to the *total hectare area farmed* per Ward.

Cattle represented the total number of *cattle* in each Ward.

Pigs represented the total number of *pigs* per Ward.

Poultry represented the total number of *poultry* in each Ward.

The livestock variables (*cattle*, *pigs*, and *poultry*) were rescaled from large continuous variables into four-level categorical variables; representing none, small, medium and large quantities of each animal per Ward.

2.3 | Statistical analysis

Mplus (Version 8.3) was used for the analysis.⁴¹ To avoid convergence problems, three predictor variables were rescaled; living alone and farming profit were divided by 10 while deprivation was divided by 100 in order to rescale their large variances. This study's integrated path model postulated new ways in which farming intensity scores, farming profit and socio-demographic variables may influence Ward mortality risk within 2001 and 2011 groups. The *standardised* results have been reported throughout this paper in order to compare the strength of relationships between paths.⁴²

Path analyses was used to assess whether mediation pathways (operating through endogenous variables) strengthen the relationships between exogenous variables to mortality. Path analysis was utilised as it can calculate the *direct*, *indirect*, *sum of indirect* and *total effects* together within the same model at 2001 and 2011.^{28,29}

Multi-group analysis was performed to cross-validate whether the structure of the path model was performing equally across groups (years) within the pooled data set.³³

Modifications from the modification indices were only incorporated into the constrained model if they produced a statistically preferable model without altering the research questions.

3 | RESULTS

3.1 | Descriptive statistics

Table 1 presents the differences between the national summary statistics in 2001 and 2011 relating to each continuous variable utilised in the model.^{5,38} Between the population censuses, Wards experienced an increase in residents *living alone* (19%), *aged over 65 years old* (18%), *unpaid carers* (16%), *LLTIs* (9%), *overall population* (8%), and *full-time workers* (1%); alongside a decrease in those with *below degree* qualifications (2%). Over this decade, farm census results indicated there was a simultaneous downturn by 2011 in the number of *farms* (18%), *farmers* (17%), *grass-based farming* (8%) and *cattle* (5%); yet, an upsurge was observed across *poultry* (36.8%), *annual farm*

TABLE 1 Summary statistics for mortality, socio-demographic, economic and environmental characteristics between 2001 and 2011 censuses.

Variables	2001	2011	% diff.
NI population	1 685 267	1 810 863	+7.45
Mortality	14 513	14 204	-2.13
65 to 100+ years	223 325	263 720	+18.09
Males	821 449	887 323	+8.02
Living alone	629 128	749 466	+19.13
Below degree	1 497 756	1 472 319	-1.70
Farmers	56 362	46 848	-16.88
Full-time hours	533 394	540 224	+1.28
Unpaid carers	185 066	213 980	+15.62
LLTI	343 107	374 646	+9.19
Farms	29 817	24 436	-18.05
Grass	993 966	917 986	-7.64
Cattle	1 679 132	1 590 452	-5.28
Pigs	385 559	425 268	+10.30
Poultry	14 347 483	19 622 510	+36.77
Annual farm profit	7984	10 497	+30.04

Note: Diff. represents the percentage difference between census years. LLTI represents limiting long-term illness. Farm profit represents 1-unit which is equal to €1000. Deprivation (NIMDM) not included in table as this is an area ranking variable.

profit (30%) and pigs (10%). Concurrent to these fluctuations, the total number of *all-cause mortalities* reduced by 2% in 2011.

3.2 | Path model fit

The seven exogenous models to mortality varied in their level of complexity; as reflected in the number of their direct and indirect pathways within each equation (Appendices S1 and S2).

The 2001 model fit results were as follows: χ^2 (15, $N=582$) = 49.825, $p < 0.001$, CFI = 0.988, RMSEA = 0.063 (90% CI [0.044–0.083]), SRMR = 0.024. The 2011 model fit results were: χ^2 (15, $N=582$) = 48.608, $p < 0.001$, CFI = 0.988, RMSEA = 0.062 (90% CI [0.043–0.082]), SRMR = 0.020. Two modification indices (MI's) were added to the 2001 model based on previous research and theoretical justification to obtain a superior model fit (LLTIs ON natural log of the population; full-time hours WITH unpaid care). The model accounted for significant variance in: *mortality* (R^2 2001 = 0.66, 2011 = 0.65); *LLTIs* (R^2 2001 = 0.85, 2011 = 0.87); *full-time hours* (R^2 2001 = 0.39, 2011 = 0.15); *unpaid carers* (R^2 2001 = 0.51, 2011 = 0.40); and *farming intensity scores* (R^2 2001 = 0.62, 2011 = 0.65).

3.3 | Direct effects

The results indicated that *natural log of the population* and *age* had a statistically significant positive association with mortality risk in Wards both years. In contrast, *below degree* had a statistically significant negative association with mortality risk in Wards both years. Furthermore, *farming profit* had a non-significant positive (2001) and non-significant negative (2011) association with mortality risk in Wards.

3.4 | Sum of indirect effects and total effects

Table 2 results indicated that *natural log of the population's* sum of indirect effects had a positive but non-significant association with *mortality* risk in Wards both years. Nevertheless, *natural log of the population's* total effect had a statistically significant positive association with mortality risk in Wards both years. *Age's* sum of indirect and total effects had a statistically significant positive association with *mortality* risk in Wards both years. The sum of indirect and total effects for *living alone* obtained identical results each year; statistically significant and positively associated with *mortality* risk in Wards both years. The sum of indirect effects for *below degree* yielded a statistically significant positive association with *mortality* risk in Wards both years. *Below degree's* total effects had a statistically significant negative association with *mortality* in 2011 Wards and non-significant negative association with *mortality* in 2001 Wards.

Farming profit's sum of indirect effects had a negative association with *mortality* risk in Wards both years; statistically significant in 2001 but non-significant in 2011. Its total effects had a positive association with *mortality* risk in 2001 Wards and negative association with *mortality* risk in 2011 Wards, however coefficients were non-significant both years. The sum of indirect effects and total effects obtained identical results both years for *males*; indicating a non-significant negative association with *mortality* risk in 2001 Wards and statistically significant positive association with *mortality* risk in 2011 Wards. The sum of indirect effects and total effects obtained identical results each year for *deprivation*; indicating a statistically significant negative association with *mortality* risk in Wards both years.

3.5 | Multi-group model

The unconstrained model fit results were: χ^2 (30, $N=1164$) = 98.433, $p < 0.001$; χ^2_{groups} = (2001) 49.825, (2011) 48.608; CFI = 0.988; RMSEA = 0.063 (90% CI

TABLE 2 Standardised results for direct, sum of indirect, and total effects in 2001 and 2011 data sets ($n = 582$ wards).

Paths to mortality	Effects	2001				2011			
		β	SE	Est./SE	p-Value	β	SE	Est./SE	p-Value
Natural log of the population	Direct	0.643	0.029	22.260	0.000	0.746	0.032	23.191	0.000
	Sum of indirect effects	0.012	0.010	1.171	0.242	0.024	0.013	1.848	0.065
	Total effect	0.655	0.029	22.937	0.000	0.770	0.031	24.896	0.000
Age	Direct	0.363	0.037	9.792	0.000	0.376	0.042	8.994	0.000
	Sum of indirect effects	0.166	0.026	6.465	0.000	0.217	0.030	7.139	0.000
	Total effect	0.529	0.028	18.873	0.000	0.593	0.032	18.567	0.000
Below degree	Direct	-0.157	0.039	-4.047	0.000	-0.190	0.044	-4.270	0.000
	Sum of indirect effects	0.110	0.015	7.426	0.000	0.114	0.019	6.148	0.000
	Total effect	-0.047	0.032	-1.440	0.150	-0.076	0.037	-2.037	0.042
Farming profit	Direct	0.018	0.026	0.691	0.490	-0.041	0.027	-1.558	0.119
	Sum of indirect effects	-0.014	0.003	-4.167	0.000	-0.001	0.003	-0.311	0.756
	Total effect	0.004	0.026	0.160	0.873	-0.042	0.026	-1.606	0.108
Males	Sum of indirect effects	-0.001	0.014	-0.066	0.948	0.025	0.010	2.592	0.010
	Total effect	-0.001	0.014	-0.066	0.948	0.025	0.010	2.592	0.010
Living alone	Sum of indirect effects	0.093	0.021	4.350	0.000	0.141	0.019	7.251	0.000
	Total effect	0.093	0.021	4.350	0.000	0.141	0.019	7.251	0.000
Deprivation	Sum of indirect effects	-0.162	0.024	-6.754	0.000	-0.206	0.026	-8.077	0.000
	Total effect	-0.162	0.024	-6.754	0.000	-0.206	0.026	-8.077	0.000

Note: Values in bold represent $p < 0.05$. Deprivation represents the Rank of Northern Ireland Multiple Deprivation Measure (NIMDM). The results for direct and specific indirect relationships are reported within the [supplementary materials](#).

[0.049–0.077]); and SRMR=0.022. The *general* analysis type was used within the pooled data set. All the parameters of the constrained model were held constant apart from three statements; in accordance with the modification indices and findings of previous literature (farming intensity score ON farming profit; full-time hours ON LLTIs; and full-time hours WITH unpaid care). The constrained model fit results were: χ^2 (57, $N=1164$)=184.121, $p<0.001$; $\chi^2_{\text{groups}}=(2001) 79.007, (2011) 105.114$; CFI=0.978; RMSEA=0.062 (90% CI [0.052–0.072]); and SRMR=0.046.

4 | DISCUSSION

4.1 | Path model

This conceptual model was built from an amalgamation of substantive theory and previously identified strong predictors of mortality. In keeping with much governmental level data, these data were de-identified and aggregated at Ward level; 2001 ($n=582$) and 2011 ($n=582$). Resizing three variables provided meaningful estimates for all model parameters and ensured the variances/covariances were fixed between 1 and 10, making interpretation easier without losing information or distorting the results. This geographical level of analysis has the advantage that it can be linked with other data at the same level and has been shown to be very effective at identifying clusters such as cancers⁴³ and suicides.⁴⁴

4.2 | Direct, sum of indirect effects, and total effects

4.2.1 | NLpop $\rightarrow$ mortality

It was unsurprising that *natural log of the population* was the strongest direct predictor of mortality both years, as this variable was derived from the size of the Ward population and represented the area's exposure rate.⁴⁵ Furthermore, NI population increased by 7% between 2001 and 2011, which also increased risk of mortality. Current results supported previous research, demonstrating that *natural log of the population to deaths* operating through *LLTIs, unpaid carers*, and *full-time hours* contributes to our understanding of mortality risk at Ward level both years by providing evidence of: (i) direct effects and (ii) total effects.

4.2.2 | Age $\rightarrow$ mortality

As surmised, results indicated Wards with higher percentages of individuals aged *over 65years old* directly

increased the risk of *mortality* compared to Wards with higher proportions of residents *under 65years old* in both cohorts. The sum of indirect and total effect results for *age* operating through four mediating variables to mortality was predictive of mortality risk in 2001 and 2011. This is consistent with previous mortality studies concluding that higher numbers of residents *over 65years old* increases the Ward level of mortality.¹¹ In addition, NI summary statistics relating to the total mortality reported 87% in 2001 ($n=12\,620$) and 80% in 2011 ($n=11\,348$) of mortality occurred in those aged over 65years old.³⁸ Current results supported previous research, but findings also demonstrated that *age to mortality* operated through *LLTIs, unpaid carers, full-time hours* and *farming intensity scores*. The findings of this study helped explain some of the variation in risk of *mortality* at Ward level both years by providing evidence of: (i) direct effects, (ii) summed indirect effects and (iii) total effects.

4.2.3 | Below degree $\rightarrow$ mortality

When controlling for the other variables in the model, the direct effect indicated that on average Wards with below degree qualifications had a lower mortality rate. These direct effects (both years) and the total effects (2011) were unexpected, but this must be seen in the context of a series of indirect effects. This unexpected result may be the consequence of migrant workers without degrees being heavily concentrated within a small number of Wards. The sum of indirect effects for Wards with a higher proportion of individuals with a *below degree* qualification had a direct effect on all the endogenous variables in the model, including mortality, and these effects were predictive of mortality risk in both 2001 and 2011. The sum of indirect effect results were in line with previous research concluding that low levels of health literacy may: (i) result in one not recognising or managing an LLTI; or (ii) it may increase their risk of developing an LLTI due to lack of health and safety knowledge (e.g., cancer, diabetes and poisoned.), thus leading to premature mortality.^{46,47}

Current results were not as posited possibly due to 65% of NI population living in urban areas, below degree qualifications decreased by 2% in 2011, and other contextual factors may have contributed to the risk of mortality within urban Wards both years. Future studies should obtain records for length of time farming and previous occupations/education at Ward level within/between 2001 and 2011 farmer status which may help explain current results. Current findings providing evidence of summed indirect effects and results supported previous research, but findings also demonstrated that *below degree to mortality* operating through *LLTIs, unpaid carers*, and *full-time hours*

helped explain some of the variation in risk of mortality at Ward level both years.

4.2.4 | Farming profit → mortality

The sum of indirect effects revealed high amounts of *farming profit* in Wards was protective against mortality risk in 2001 only. The non-significant results in 2011 may be a by-product of the decrease in the number of farms (18%) from 2001 to 2011 in NI and farm diversification strategies (members of farming households' need to supplement their on-farm income via off-farm employment).^{14,48} Findings also demonstrated that *farming profit to mortality* operating through *LLTIs*, *unpaid carers*, *full-time hours*, and *farming intensity scores* helped explain some of the variation in risk of death at Ward level both years by providing evidence of summed indirect effects in 2001.

4.2.5 | Males → mortality

In 2011, the total effects provided support for the hypothesis as *male* was a statistically significant predictor of a greater number of mortality at Ward level. Current 2011 results supported previous research.^{8,9} In 2011, the sum of indirect and total effects results provided support for their hypothesis as *male* was a statistically significant predictor of risk of mortality. The 2011 findings helped explain some of the variation in risk of mortality at Ward level, by providing evidence of: (i) summed indirect effects and (ii) total effects.

4.2.6 | Alone → mortality

The sum of indirect effects and total effect results provided support for their hypothesis as *living alone* was a statistically significant predictor of mortality risk both years. Current results supported previous research³³ and may have been obtained as *living alone* increased by 19% from 2001 to 2011 in NI. Current results may have been obtained as *living alone* increased by 19% from 2001 to 2011 in NI. Findings also demonstrated that *living alone to mortality* operating through *LLTIs*, *unpaid carers*, *full-time hours* and *farming intensity scores* helped explain some of the variation in risk of mortality at Ward level both years by providing evidence of: (i) summed indirect effects and (ii) total effects.

4.2.7 | NIMDM → mortality

The sum of indirect and total effect results indicated that Wards ranked as least deprived were protective against

mortality risk both years. In other words, results supported their hypothesis as Wards ranked as most deprived were predictive of mortality risk both years. Results indicated the most deprived Wards were predictive of mortality operating through four endogenous variables which match those observed in previous research.^{8,17,46,49} Findings also demonstrated that *deprivation to mortality* operating through *LLTIs*, *unpaid carers*, *full-time hours*, and *farming intensity scores* helped explain some of the variation in risk of mortality at Ward level both years by providing evidence of: (i) summed indirect effects and (ii) total effects.

4.3 | Strengths and limitations

This type of multi-source data analyses with new research questions, induced the detection of new mortality trends that may otherwise have remained undiscovered. As posited, this study demonstrates that the nature of the relationship between areas *farming intensity scores*, *farming profit*, and *socio-demographics* on *mortality* risk was dependent on the specific social-environmental characteristics of Wards studied. According to Kane et al.,⁵⁰ there are some limitations when using secondary data; such as a lack of exact specificity and the choice of indicators is dictated by the data available. Area-level studies using aggregated data suffer from a degree of unavoidable ecological bias. Researchers must be cautious not to make an ecological fallacy when drawing inferences from area-level results.⁵¹ This danger has made some researchers reluctant to analyse clustered or aggregated data as there is a misconception it has restricted use and power.⁵²

5 | CONCLUSION

This study challenges the misconception that an agricultural areas populace is healthier than non-agricultural area dwellers by investigating the geographical variation and spatial trends of mortality rates within NI Wards.^{3,4,53,54} Overall results provided evidence for exogenous variables to mortality both years via their: (i) *direct effects* (age), (ii) *summed indirect effects* (age; males, [2011]; living alone; farming profit, [2001]; deprivation), and (iii) *total effects* (age; males, [2011]; living alone; deprivation). The multi-group findings provided cross-validation of these new conceptual cause-and-effect relationships relating to mortality within the pooled data set.⁵² In line with earlier area level research, results support the aggregated interpretation that higher levels of farming activity within a Ward increases the risk of poor health and mortality within those Wards.

Results supported the overarching hypothesis indicating that agriculturally saturated Wards have higher

mortality risk in comparison to non-agriculturally based Wards at single time points. This study was able to describe the strength and confirm the independent importance of farm census variables in explaining mortality risk variation within Wards. Results provide a novel contribution as few studies have simultaneously modelled farming profit, farming intensity scores, unpaid care, and socio-demographic attributes on mortality risk at Ward level. In line with earlier area level research, results support the aggregated interpretation that higher levels of farming activity within a Ward increases the risk of poor health and mortality within those Wards. To the authors knowledge, this is the first study of its kind within NI.

This study's farm census to population census linkage has provided a contextual mechanism for investigating and understanding some of the unexplained variations in mortality risk at Ward-level in NI, allowing future inter-disciplinary researchers to make international comparisons.^{5,12} Future research in this area is necessary, the authors plan an expansion of this study with the addition of the 2021 Census information. The overall model will be tested cross-sectionally at 2021 and longitudinally between the three time points (2001, 2010, 2021) once the information becomes available via NINIS online. This was an essential study to enable tailoring of new strategies and upgrading current policies to bring about significant change at local level (to help reduce on-farm injuries and deaths).

AUTHOR CONTRIBUTIONS

Kelly Trearty: Conceptualization; investigation; writing – original draft; methodology; validation; visualization; writing – review and editing; formal analysis; project administration; data curation; resources; software. **Brendan Bunting:** Resources; validation; software; methodology; investigation; supervision; project administration; writing – review and editing; writing – original draft; visualization; formal analysis; data curation; conceptualization. **John Mallett:** Resources; validation; software; writing – review and editing; project administration; supervision; investigation; methodology; writing – original draft; visualization; formal analysis; data curation; conceptualization.

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CONFLICT OF INTEREST STATEMENT

There were no competing interests or conflict of interests during this research.

DATA AVAILABILITY STATEMENT

These data were publicly available from NINIS. The Mplus input and output analysis files are available upon request from Dr. Kelly Trearty.

ETHICS STATEMENT

Ethics approval was granted by Ulster University Research Ethics Committee in January 2019 (Reference ID: B00573168).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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